

Water Balance of a Managed Reservoir under Climate Change: Complementary Process-Based and Lumped Conceptual Modeling

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A Context

Drinking-water reservoirs face **stronger climate variability** and **anthropogenic fluxes** such as **withdrawals** and **spillway restitutions**, which make the balance **nonstationary**. Many studies^[1] close the balance with a single residual that mixes evaporation, groundwater inflows, and **under-dam leakage**. Because sites and data differ, **modeling must be site specific** and combine **process-based distributed models** that **resolve mechanisms** but need **rich data** with **lumped models** that **run fast** on sparse data and keep a **physical anchor**.

Our choice: use **two complementary methods** to build a clear and transferable water balance.

- a **process-based distributed model** to **separate mechanisms** and **map exchanges** between reservoir, aquifer, and river under operations. This model **revealed and quantified under-dam leakage**;
- a **lumped conceptual Bayesian (budget) model** to **estimate aggregated fluxes and their uncertainty**. It is **validated against the process-based model**, used as a **surrogate** for rapid scenarios, and **stand-alone** on data-scarce sites.

Problem Statement:

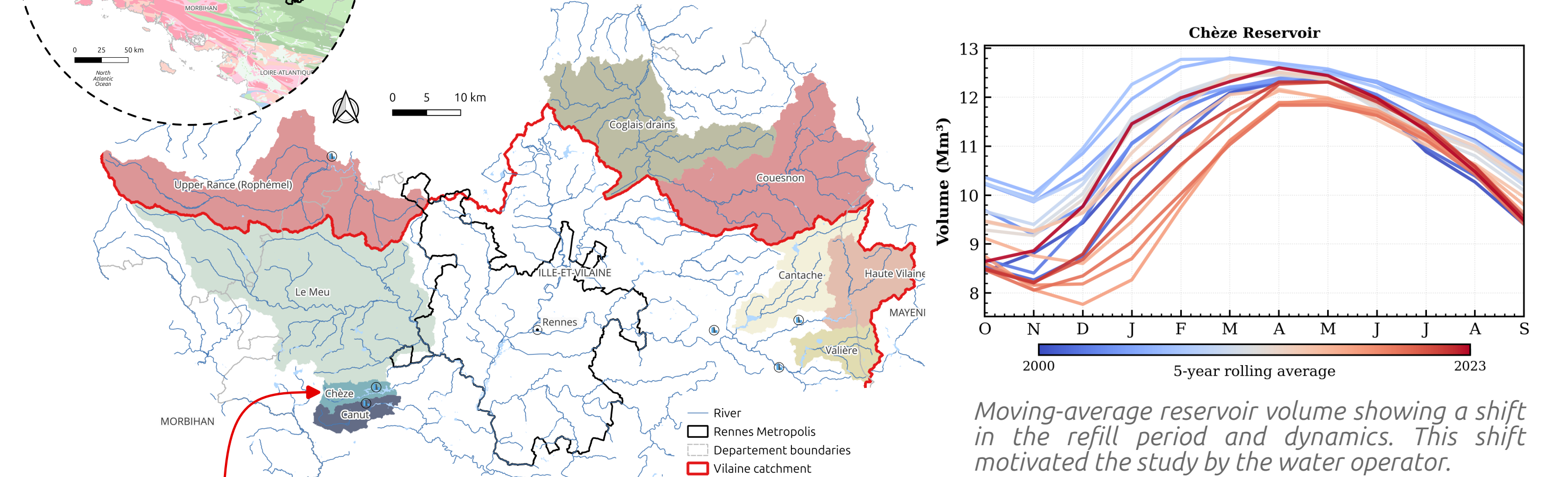
- How to build a **robust and transferable water balance** for a **managed reservoir**, with **clear flux attribution** (including under-dam leakage) and **quantified uncertainty**, by combining a **process-based model** with a **lumped conceptual model** under present and future climate forcing?

Key Contributions & Objectives

- **Explicit under-dam leakage** in the water balance (not a residual) and **clear flux attribution**.
- **Two complementary models:** process-based (mechanisms, exchange maps under operations) + lumped Bayesian (aggregated fluxes + uncertainty, surrogate & stand-alone).
- **Common evaluation metrics** for both models: goodness-of-fit, **mass-balance closure**, and **uncertainty indices** → results are comparable.
- **Validate** the lumped model against the process-based reference, then **use it as a surrogate** for fast scenario testing.
- **Transferable multi-site workflow** aimed at **operational planning** under current and future climate.

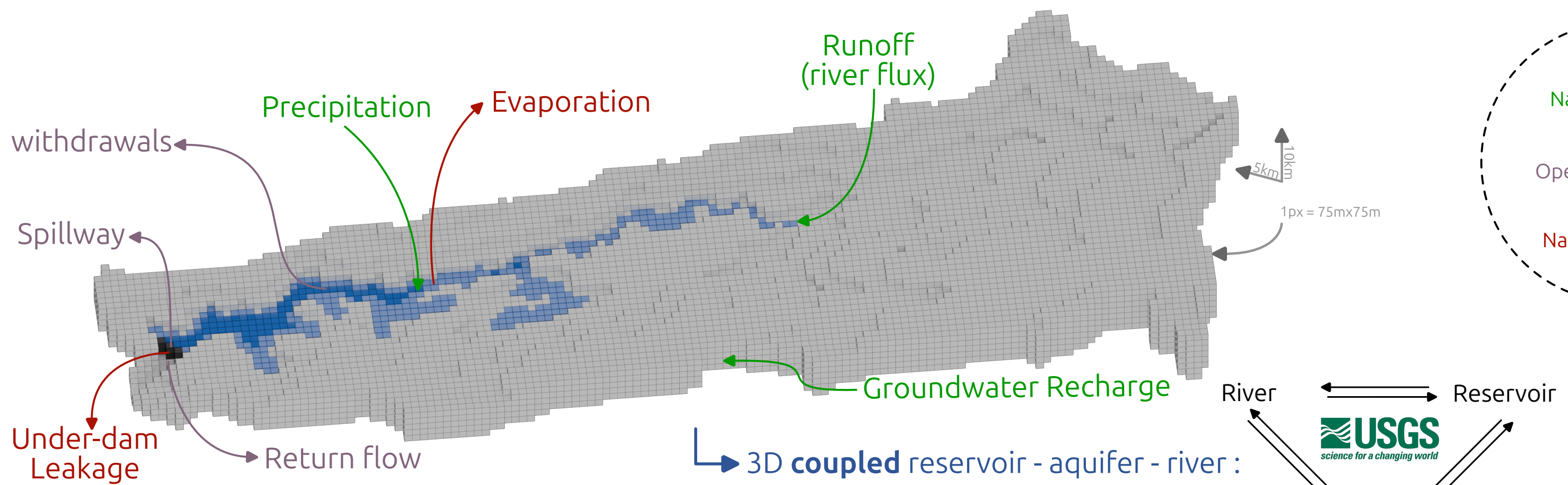
B Study Area & Sites

→ We work with Eau du Bassin Rennais (EBR) to assess the future of drinking-water resources and test if current operating practices remain viable under climate change. EBR produces **26 Mm³/yr** for **~500,000 people**, in Rennes, **~60%** comes from two reservoirs, **~30%** from surface drains, and **~10%** from dispersed wellfields, while nearby towns rely on **three additional reservoirs** outside EBR operations that we also study.



- **Chêze signal**: **delayed refill**, **lower seasonal peaks**, **chronic recharge deficit** → motivation for this thesis.^[2]
- Study focus: small **dammed catchments** in **basement** (hard-rock) geology in **Brittany, France**
- Pilot site: **Chêze-Canut** with rich data (bathymetry, operations, daily SIM2) enables **explicit flux separation** in a process-based model.
- **Generalization** to other reservoirs and intakes with sparse data uses a **global Bayesian model** to estimate aggregated fluxes and uncertainty and to act as a **surrogate** for rapid scenarios.

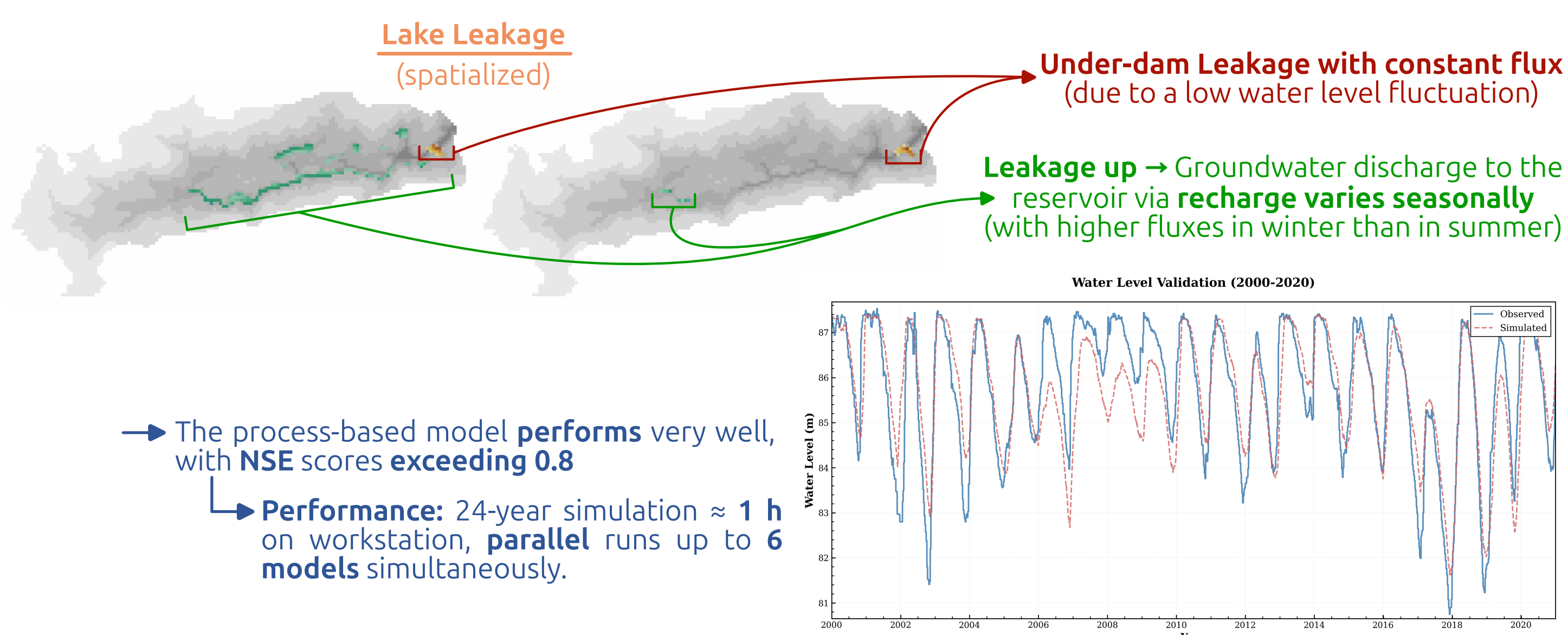
C Distributed Process-Based Model



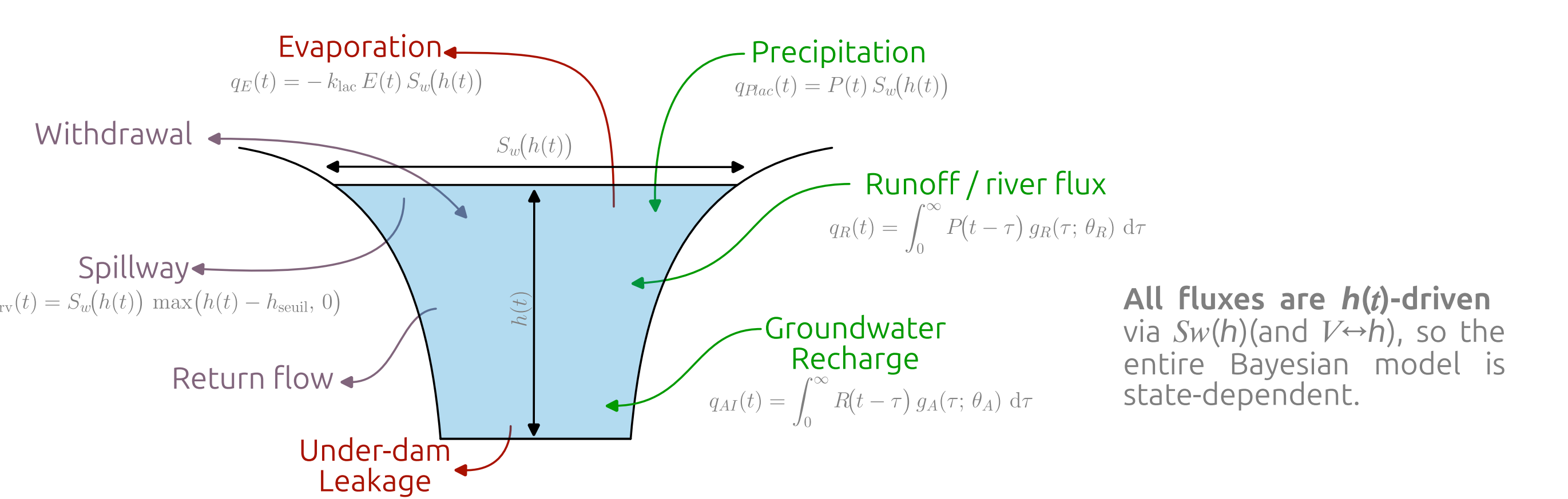
Spatially distributed model (Darcy flow + mass conservation) with daily transient simulations on a **gridded aquifer**. Calibration focused on **K (m/day)** and **Sy (%)**. Only two of MODFLOW's 100+ parameters.

- **Engine:** **MODFLOW-NWT** with **LAK** (lake) + **SFR** (stream) → explicit reservoir-aquifer-river coupling.
- **Why spatial matters:** spatial processes → **add obstacles/liners**, impose **heterogeneity maps**, move boundaries → test **design/management scenarios** and climate impacts.
- **Data demand:** needs high-resolution bathymetry, geology, forcings and **heavy data preparation**.
- **Supporting tools:** **HydroModPy**^[3] simplifies **river-reservoir-aquifer coupling** and workflows, eases setup and maintenance.

HydroModPy

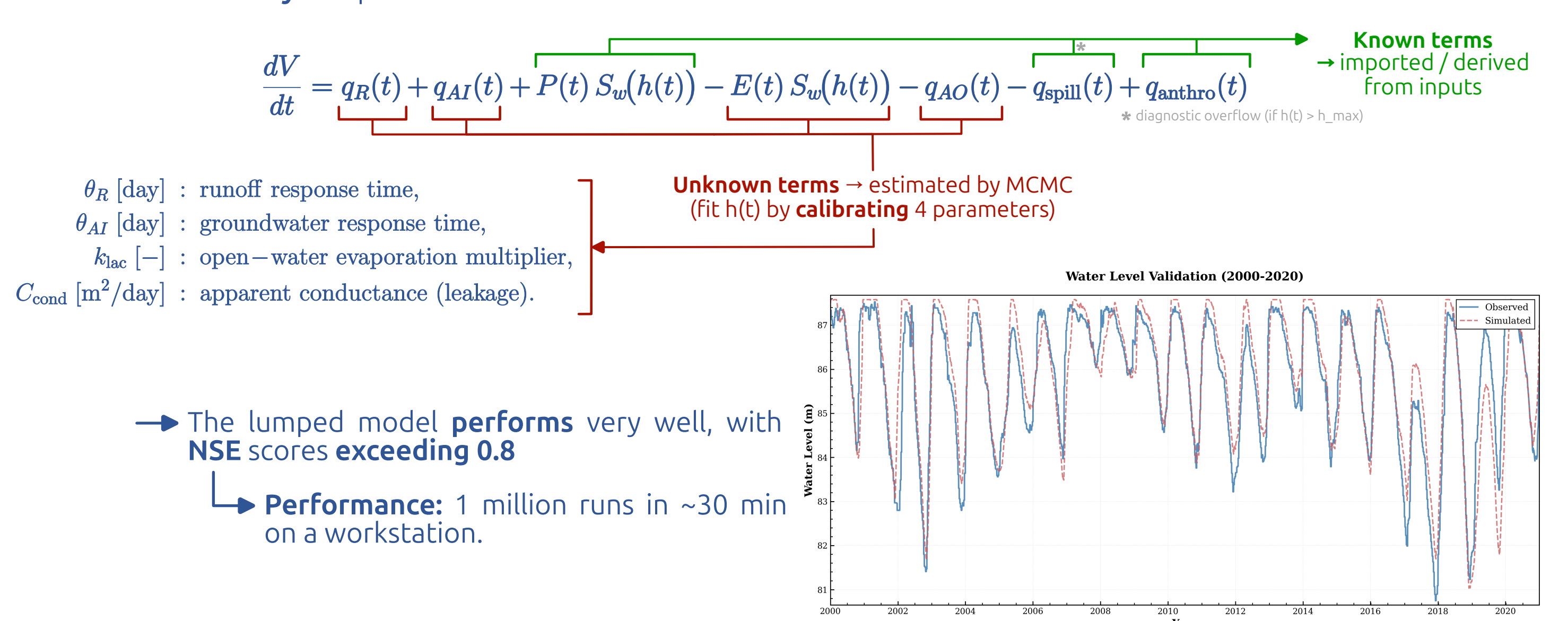


D Lumped Bayesian Budget Model



Concept (signal → Fluxes): interpret lake level **h(t)** as the **convolution of inflow/outflow signals** and use **MCMC to deconvolve them** → **each water-balance term is computed explicitly, not as a residual**.

- **Model box:** Non-spatial (lumped) reservoir, a **bathymetry abacus** maps **h(t) → V(h)** and **S_w(h)** so the model remains physically consistent.
- **Sampler:** Metropolis-Hastings with adaptive Gaussian proposals.
- **Likelihood:** error on level/volume time series (e.g., Gaussian; NSE/ RMSE-based).
- **Data:** only **time series** (level, P/ET, optional operational flows), **no heavy data preparation**.
- The model was developed **entirely in Python from scratch**, providing **improved maintainability** and **usability** compared to MODFLOW.



E Summary : When to Use Which

- MODFLOW** when you need **spatial** answers: maps of lake-aquifer exchange, sensitivity to **heterogeneity**, and the impact of **design/operation** changes (liners, barriers, rules). It **localizes processes**, delivers **where/why** diagnostics, and **evaluates** engineering options under stress scenarios.
- Bayesian lumped** when you need **speed** and **sparse data**. Deploy **quickly**, cover **many sites**, and quantify **uncertainty** on levels and fluxes, with **lightweight calibration**, **minimal preprocessing**, and easy **scenario sweeps** for screening.
- As a **surrogate**, the Bayesian model **screens scenarios** and projections, then provides priors and initial targets to **MODFLOW**, yielding **fewer heavy runs**, **faster convergence**, and **physics-consistent** results with transparent trade-offs.

F Perspectives

Surface model for recharge - runoff

- **Why?** → Replace unusable generic recharge with a **physically consistent surface model** usable at site scale and for projections.
- **Scenarios** → Vary **land cover**, soils, and **management** to test climate-change and operational options.
- **Forcings** → Plug-and-play with ERA5-Land, CERRA-Land, SIM2, ...
- **Integration** → Ongoing in **HydroModPy** (with **University of Neuchâtel**): one pipeline, reproducible, versioned.

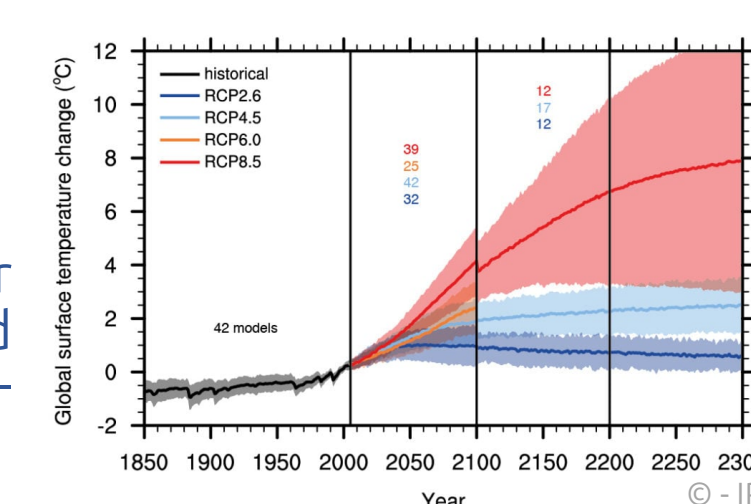
Benefits → Consistent inputs across frameworks, supports **projection studies**, reduces bias vs. generic recharge, enables rapid **scenario sweeps**.

PyHELP^[4]
A Python library for the assessment of spatially distributed groundwater recharge and hydrological components with HELP

Climate projections & operations

- **Purpose** → Stress-test reservoir **viability** under future climate, anticipate **shortages**, and design **operating rules** that safeguard drinking-water supply.
- **Workflow** → **Bayesian surrogate** sweeps thousands of (**climate × management**) scenarios to find **viable data ranges**, the best few go to **MODFLOW** for spatial diagnostics.
- **Outputs** → **Scenario-based seasonal ranges** with **uncertainty**, to guide **steady, low-risk management**.

Benefits → Scenario planning that is **fast (surrogate)**, **physically grounded (MODFLOW)**, and **uncertainty-aware**, built for climate projections first, not as an afterthought.



Multi-Site Planning Metamodel

- **Purpose** → Coordinate all sites as a **portfolio**, maximizing **end-of-season storage** and **reliability** while respecting **demands**
- **Engine** → Compare **statistical response surfaces**, **Bayesian ensemble control**, or **learning-based policy search**, choose the **simplest**.

Benefits → Multi-objective: **maximize reliability & final storage**, **minimize shortage & spill**, consider **cost/energy**, hard constraints on **ecological flows**.

[1] - Song, J., Her, Y., & Kang, M. (2022). Estimating Reservoir Inflow and Outflow From Water Level Observations Using Expert Knowledge: Dealing With an Ill-Posed Water Balance Equation in Reservoir Management. Water Resources Research, 58(4), e2020WR028183. <https://doi.org/10.1029/2020WR028183>

[2] - Abhervé, R. (2022). Intégration du changement climatique dans la gestion de la ressource en eau: Exemple du bassin rennais [Université de Rennes]. <https://theses.fr/2022REN18067>

[3] - Gauvain, A., Abhervé, R., Coche, A., Le Mesli, M., Roques, C., Bouchez, C., Marçais, J., Leray, S., Marti, E., Figueroa, R., Bressiani, E., Vautier, C., Boivin, B., Sallou, J., Bourcier, J., Combemale, B., Brunner, P., Longuevergne, L., Aquilina, L., & De Dreuzy, J.-R. (2025). HydroModPy: A Python toolbox for deploying catchment-scale shallow groundwater models. Groundwater hydrology/Modeling approaches. <https://doi.org/10.5194/egusphere-2024-3962>

[4] - Schroeder, P. R., Aziz, N. M., Lloyd, C. M., Zapfi, P. A. (1994). The Hydrologic Evaluation of Landfill Performance (HELP) Model: Engineering Documentation for Version 3. EPA/600/R-94/168b. U.S. Environmental Protection Agency, Office of Research and Development, Washington, D.C., 126 pp. url: <https://www.epa.gov/land-research/hydrologic-evaluation-landfill-performance-help-model>.

The Bayesian lumped model offers rapid, uncertainty-aware screening, while the process-based model resolves spatial mechanisms and management options. Their combination yields a robust and transferable framework for drinking-water reservoirs under climate change.

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